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# =====  
# 19. Edge Detection under Illumination Changes  
# Google Colab Code  
# =====
```

```
import cv2  
import matplotlib.pyplot as plt  
from google.colab import files
```

```
# -----  
# STEP 1: Upload image  
# -----  
print("Upload the image")  
uploaded = files.upload()
```

```
# Get uploaded file name  
img_path = next(iter(uploaded))
```

```
# -----  
# STEP 2: Read image in grayscale  
# -----  
img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
```

```
# -----  
# STEP 3: Create illumination variations  
# -----  
# Dark image  
dark = cv2.convertScaleAbs(img, alpha=0.5, beta=0)
```

```
# Normal image  
normal = img.copy()
```

```
# Bright image  
bright = cv2.convertScaleAbs(img, alpha=1.5, beta=40)
```

```
# -----  
# STEP 4: Apply Canny Edge Detection  
# -----  
edge_dark = cv2.Canny(dark, 100, 200)  
edge_normal = cv2.Canny(normal, 100, 200)  
edge_bright = cv2.Canny(bright, 100, 200)
```

```
# -----  
# STEP 5: Display Original Images  
# -----  
plt.figure(figsize=(15,4))
```

```
images1 = [dark, normal, bright]  
titles1 = ['Dark Image', 'Normal Image', 'Bright Image']
```

```
for i in range(3):
```

```
plt.subplot(1,3,i+1)
plt.imshow(images1[i], cmap='gray')
plt.title(titles1[i], fontsize=12)
plt.axis('off')

plt.suptitle('Illumination Variations', fontsize=16, fontweight='bold')
plt.show()

# -----
# STEP 6: Display Edge Detection Results
# -----
plt.figure(figsize=(15,4))

images2 = [edge_dark, edge_normal, edge_bright]
titles2 = ['Edges - Dark', 'Edges - Normal', 'Edges - Bright']

for i in range(3):
    plt.subplot(1,3,i+1)
    plt.imshow(images2[i], cmap='gray')
    plt.title(titles2[i], fontsize=12)
    plt.axis('off')

plt.suptitle('Edge Detection under Illumination Changes',
            fontsize=16, fontweight='bold')
plt.show()
```

Upload the image

IMG 5.jpg

IMG 5.jpg(image/jpeg) - 3237854 bytes, last modified: 07/08/2026 - 100% done

Saving IMG 5.jpg to IMG 5.jpg

Illumination Variations

Dark Image



Normal Image

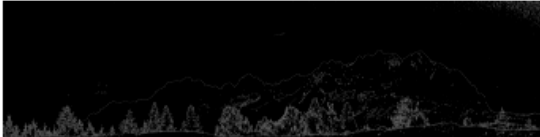


Bright Image

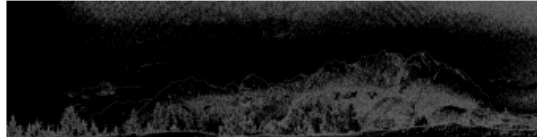


Edge Detection under Illumination Changes

Edges - Dark



Edges - Normal



Edges - Bright

